Resilient Forge H2E – Human-to-Expert Framework for Deterministic, Sovereign AI

Integrating UNESCO Resilience Pillars for Sustainable AGI

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The SOMALA Advantage

Transforms probabilistic black-box AI into verifiable intelligence.

100% on-premise deployment with zero cloud dependencies.

Guaranteed operation within strict hardware limits.

Designed for safety-critical sectors like aerospace, defense, and robotics.

The Problem – Probabilistic Uncertainty

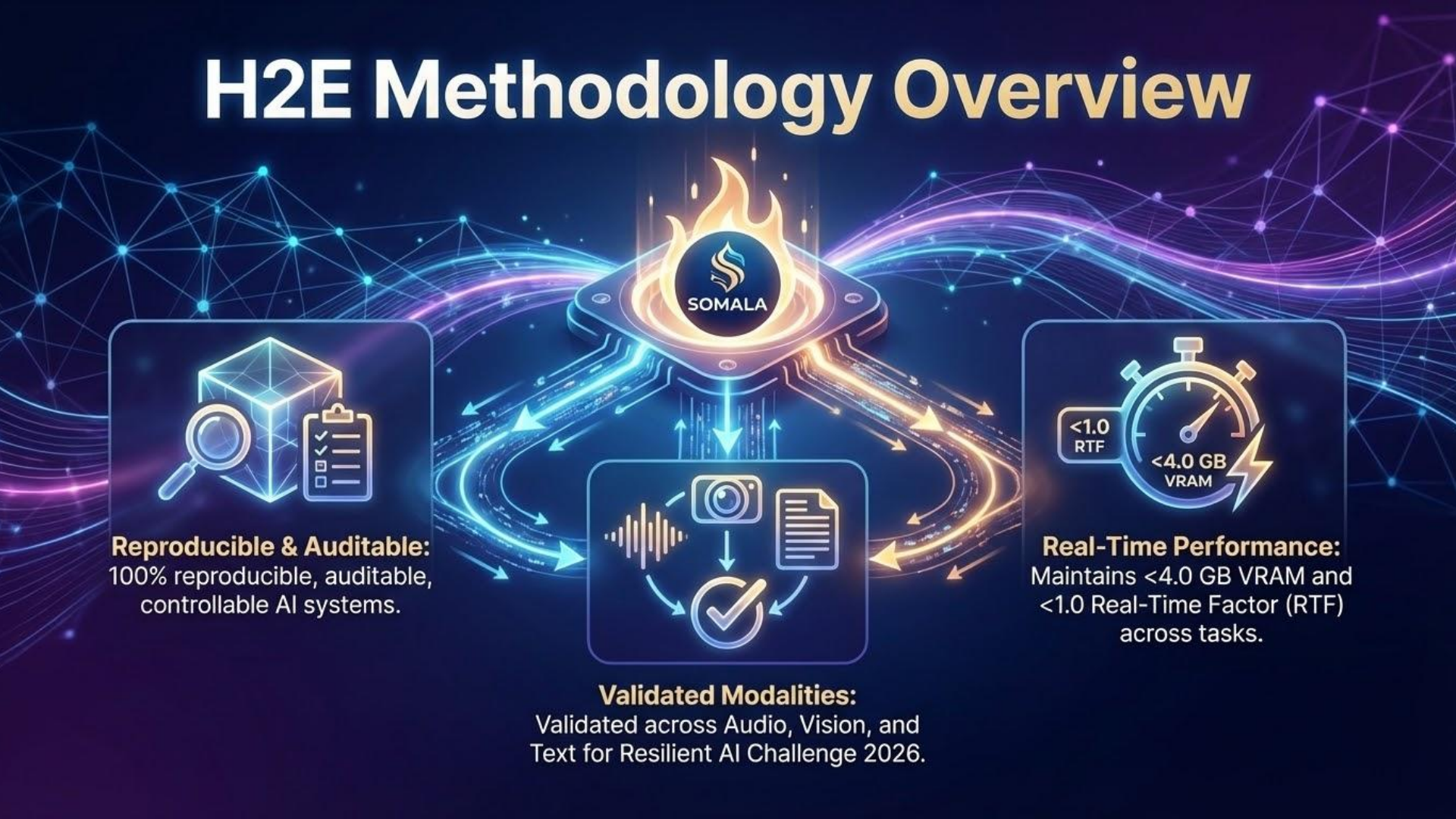
Non-Determinism: Inconsistent outputs make auditing impossible.

Memory Spikes: Unpredictable consumption leads to OOM failures.

Opaque Impact: High carbon footprints and unclear decision processes.

Vendor Lock-In: Reliance on cloud APIs compromises data sovereignty.

H2E Methodology Overview



Reproducible & Auditable:
100% reproducible, auditable, controllable AI systems.

Validated Modalities:
Validated across Audio, Vision, and Text for Resilient AI Challenge 2026.

Real-Time Performance:
Maintains <4.0 GB VRAM and <1.0 Real-Time Factor (RTF) across tasks.

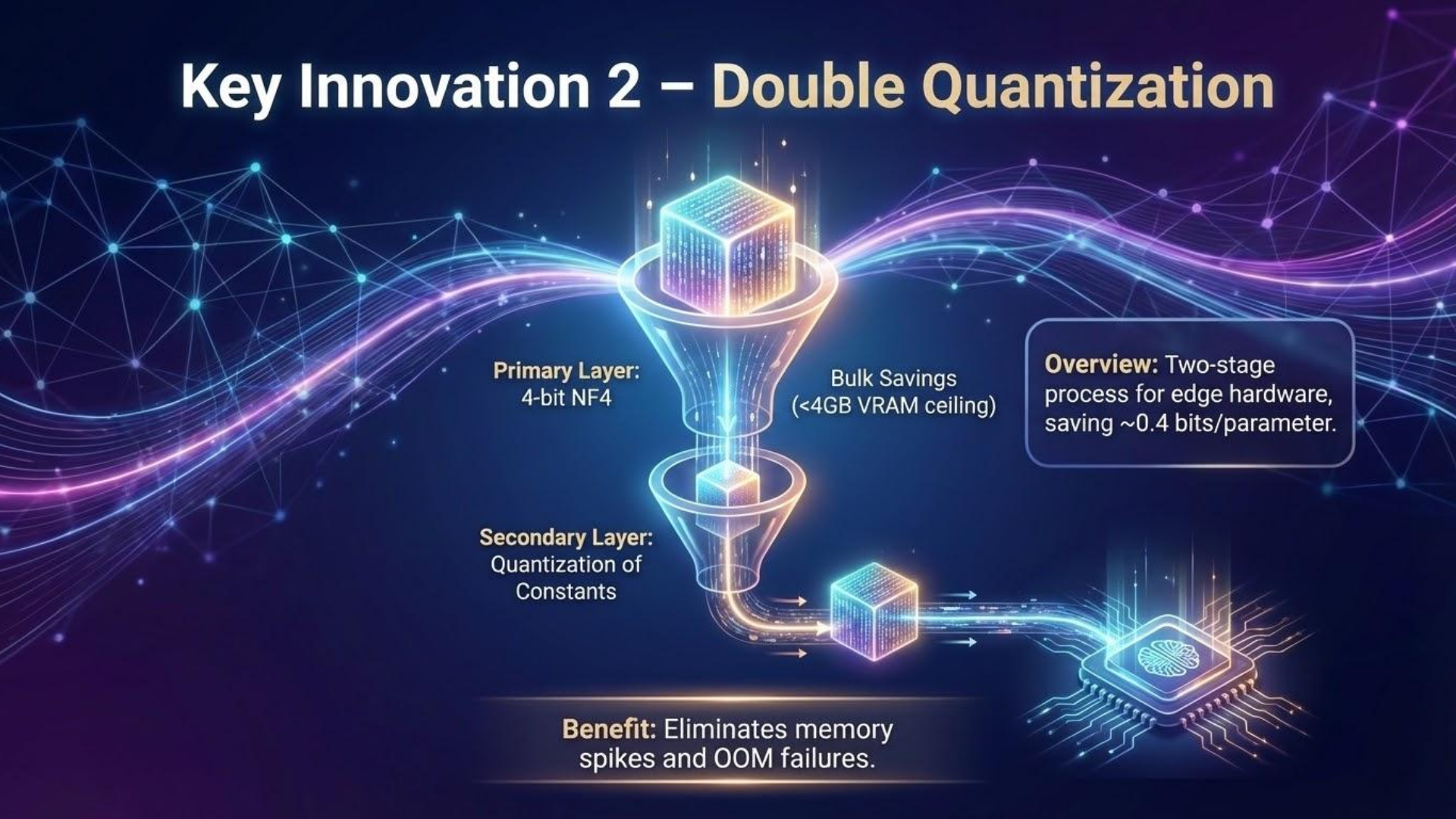
Key Innovation 1 – Determinism Lock (Seed 123)

Comprehensive Protocol:
Fixes seeds across Python, NumPy, PyTorch, and CuDNN.

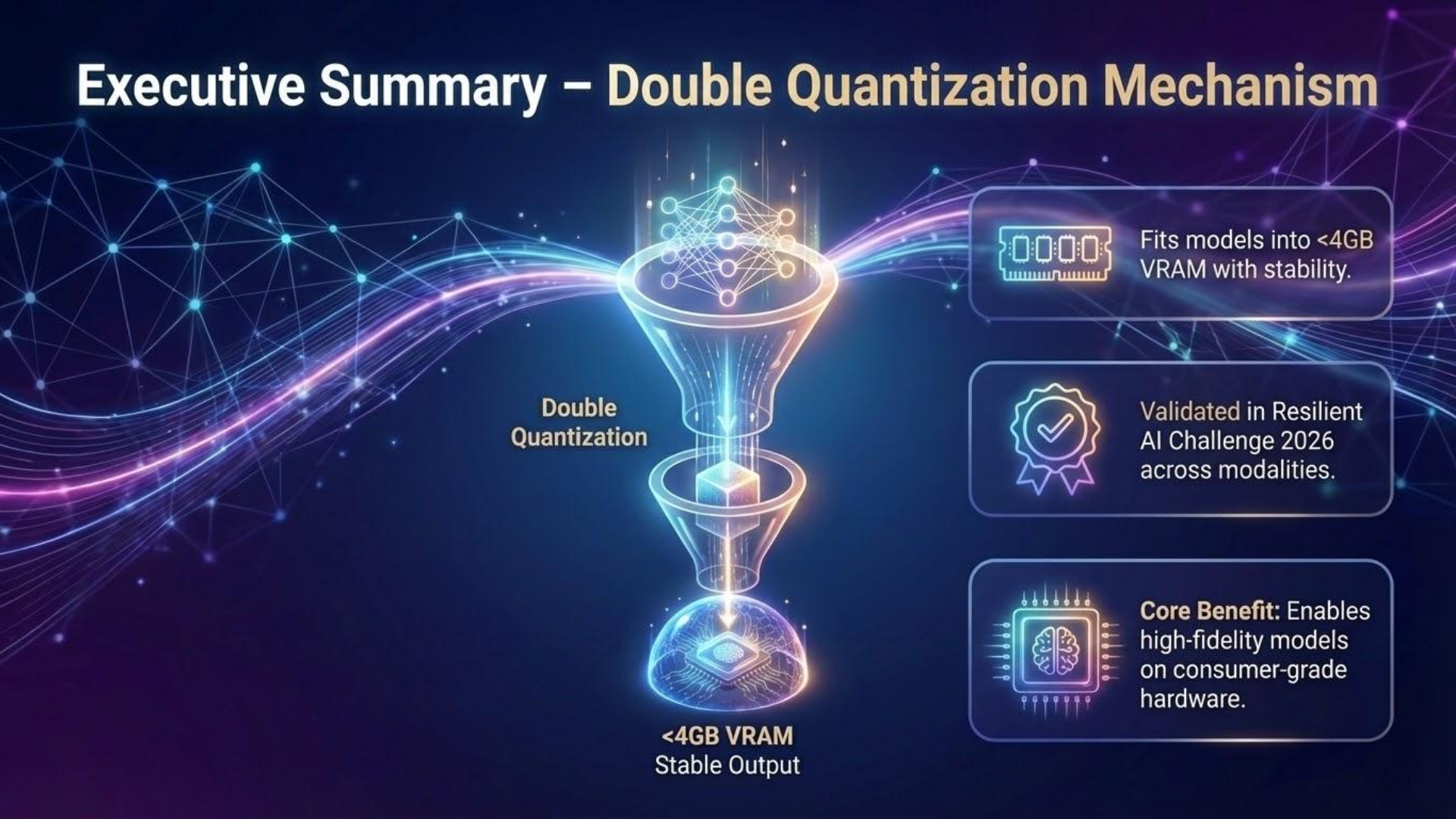
Standardization:
Uses 'Seed 123' as mandatory constant for all layers.

Outcome:
Identical outputs for identical inputs, enabling full traceability.

Key Innovation 2 – Double Quantization



Executive Summary – Double Quantization Mechanism



Validation Results – Resilient AI Challenge 2026

H2E framework demonstrates remarkable efficiency, consistently meeting <4GB VRAM and <1.0 RTF targets across diverse modalities.

Audio Processing
(Voxtral-Mini-4B)

VRAM: 2.780 GB

RTF: 0.814

Dynamic Token Allocation (70-80 tokens) for seamless handling.

Vision Tasks
(Gemma-3-4b-it)

VRAM: 3.167 GB

RTF: 0.800

Scaled Dot-Product Attention, native 224x224 resolution.

Text Generation
(Sarvam-1)

VRAM: 1.669 GB

RTF: 0.836

Reactive Response (5 tokens), Greedy Decoding for 100% deterministic output.

Double quantization turns resource-intensive models into reliable, edge-deployable tools for mission-critical applications.

Key Innovation 3 – Carbon Transparency

Real-Time Monitoring

Integrated CodeCarbon for every inference, providing instant feedback.

Verifiable Metrics

Tracks kg CO_2 per task with precision (e.g., 0.00179kg for audio).

Optimization & Impact

Enables environmentally conscious choices and sustainable AI deployment.

Transparency empowers responsible AI, balancing performance with environmental stewardship.

H2E-Concierge Framework – Architecting Deterministic AGI

Normalized Expert Zone (NEZ)

Immutable human reasoning gold standard.

Intent Governance Zone (IGZ)

Amplifies signals (12.5x multiplier), suppresses noise.

Semantic ROI (SROI)

Cosine similarity measurement with safety switch (threshold 0.9583) to prevent drift.

Biological Gap Bridge

Bridges biological gap for sustainable AGI.

Comparative Analysis – Standard AI vs. H2E-Enhanced AI

Standard AI

Behavior:
Probabilistic methods, uncertain & variable outputs, inconsistent.

Memory Management:
Unpredictable spikes, OOM risks, disrupts operations.

Deployment:
Heavy cloud dependencies, sovereignty risks, potential vendor lock-in.

H2E-Enhanced AI

Behavior:
Deterministic results, 100% identical outputs, Seed 123 protocol.

Memory Management:
Strict <4GB VRAM ceiling, complete stability, double quantization.

Deployment:
Fully sovereign on-premise setups, complete ownership & control.

Sovereign Deployment Architecture

Data Security

No data leaves premises.

Offline Capability

Operates without APIs or internet.

Audit Trail

Full traceability from model to artifact.

Target Systems

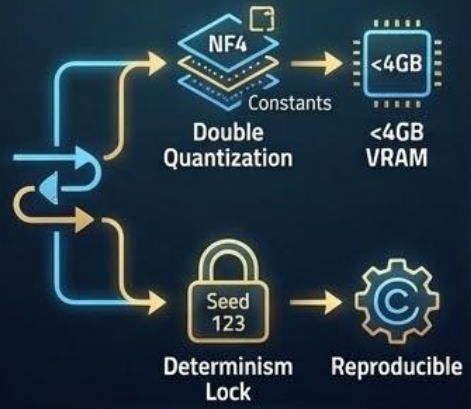
Flight control, defense, 22-DoF humanoid robots.

Submission Checklist – Resilient AI Challenge

STAGE 1: MODEL SELECTION

- Choose High-Capability Multi-Modal Models
- Establish Open-Weight Baselines
- Ready for Optimization

STAGE 2: H2E TRANSFORMATION



- Apply Double Quantization & Seed 123 Protocol
- Eliminate Variability & Memory Spikes
- Ensure Stable, Efficient Edge Deployment

STAGE 3: AUDIT REPORTS & SUBMISSION

- Generate On-Premise Audit Deliverables
- Full Transparency without Model Files
- Verify UNESCO Compliance & Sovereignty

Future Work & Roadmap

TPU Optimization

Adapt for local TPU to maximize battery endurance in 22-DoF systems.

6G Swarm Integration

Ultra-low-latency telemetry for multi-robot auditing.

Power-Aware Control

Dynamic loops adjust based on battery levels.

Formal Verification

Mathematical proofs ensure determinism remains certification-ready.

Conclusion – A New Standard for High-Integrity AI

Achieved: Deterministic AI

Converted probabilistic AI into deterministic across modalities.

System Trustworthiness

100% reproducible for audits.

Operational Efficiency

<4GB VRAM, <1.0 RTF for edge deployment.

Complete Sovereignty

Offline architecture, no vendor lock-in.

Environmental Accountability

Verifiable carbon tracking supports optimization.

Questions & Discussion

H2E Framework Details

Seed 123 for Engineering Auditability

Modality-Specific RTF Optimization

<4GB VRAM on Diverse Hardware

On-Premise, Air-Gapped Operation

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